

Large Language Models for the
Economic and Social Sciences:

Getting Output from LLMs

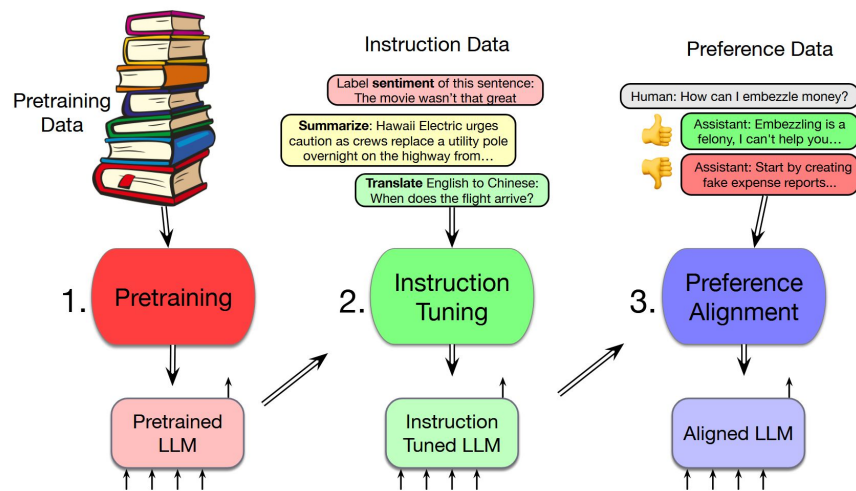
Indira Sen, HWS 2025

<https://github.com/dess-mannheim/LLM4ESS-HWS25>

Credits: Thanks to Markus Strohmaier for some of the content on prompting.

Last Time: How can we build LLMs?

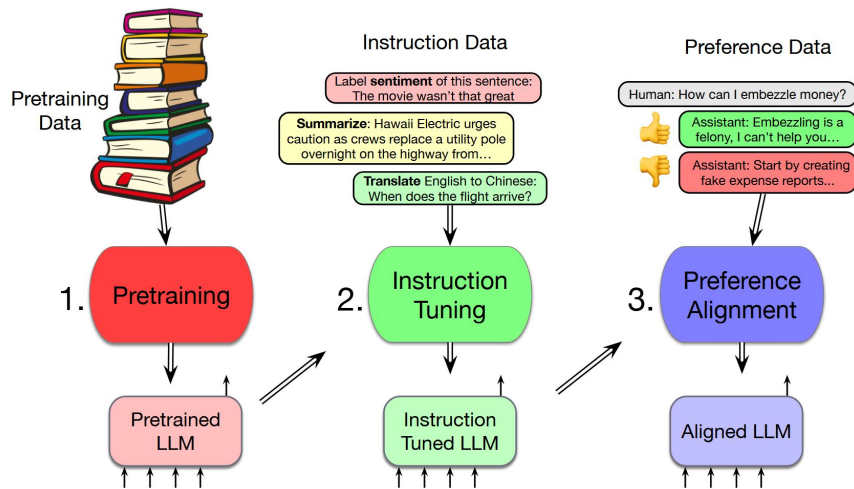
Somewhat simplified answer



Jurafsky and Martin, 2025

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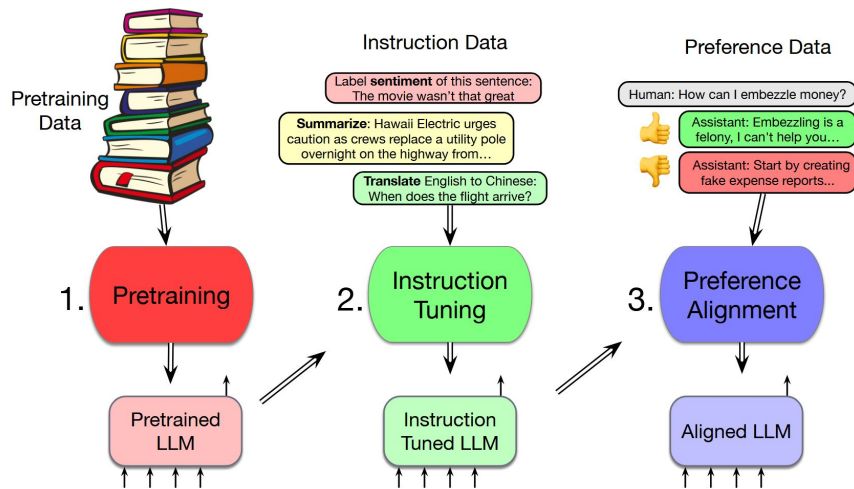
Jurafsky and Martin, 2025

Vastly Simplified answer

- We take a LOT of data encoding social, cultural, and of course linguistic traces
- Train NLP models that can generate content on this data
- We further train the model to follow and respond to instructions
- We further modify the model so that it doesn't have unwanted behavior

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Jurafsky and Martin, 2025

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- **Train NLP models that can generate content on this data**
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Summary of Last Week and Today

Last session we learned:

- Pretraining and fine-tuning
- Language generation
- Instruction-tuning
- Reinforcement Learning with Human Feedback

Today:

- Different types of prompting
- How to get outputs from LLMs in other ways
 - Structured outputs
 - Log probabilities
- Practicalities of working with LLMs
 - Which LLMs and where to find them
 - How to evaluate them

Natural Language Generation (NLG)

NLG focuses on systems that produce fluent, coherent and useful language output for human consumption

Open-ended generation: the output distribution still has high freedom

Non-open-ended generation: the input mostly determines the output generation.



How does an LLM generate content?

- Models like BERT are **encoder** only models
 - Are usually finetuned (trained on supervised data) for classification tasks.
- For generative tasks, we use either **decoder** models like GPT, LLaMa
 - It takes as input a series of tokens, and iteratively generates an output token one at a time.
 - Left to right (causal, **autoregressive**)
- Another family of generative models: **encoder-decoder** models like Flan T5, Whisper
 - Trained to map from one sequence to another
 - Useful for machine translation, speech recognition

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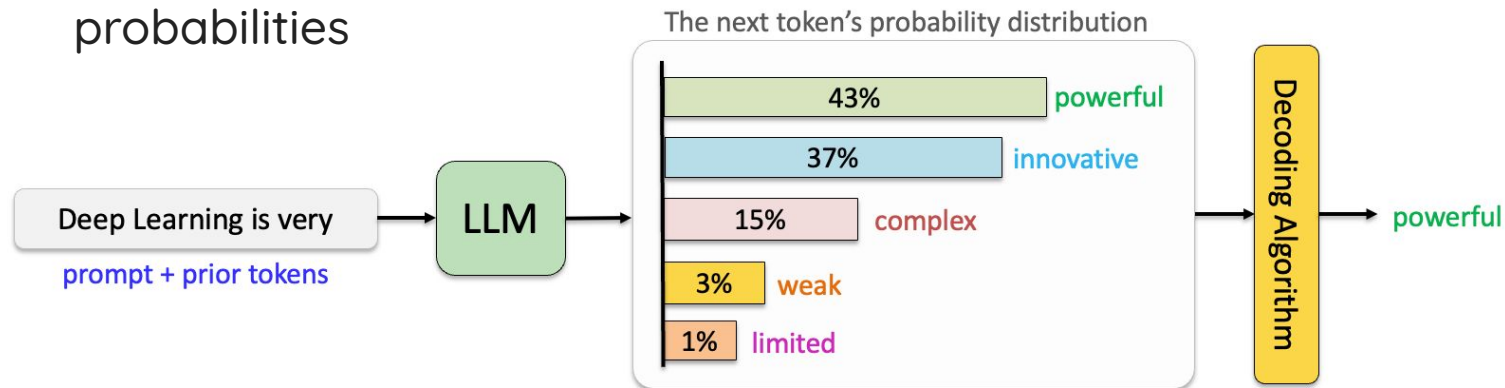
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- Conditioned on previous contexts, e.g., some type of input or **prompt**
 - computing the probability of the next token from the prior context
 - sampling from that distribution to generate a token

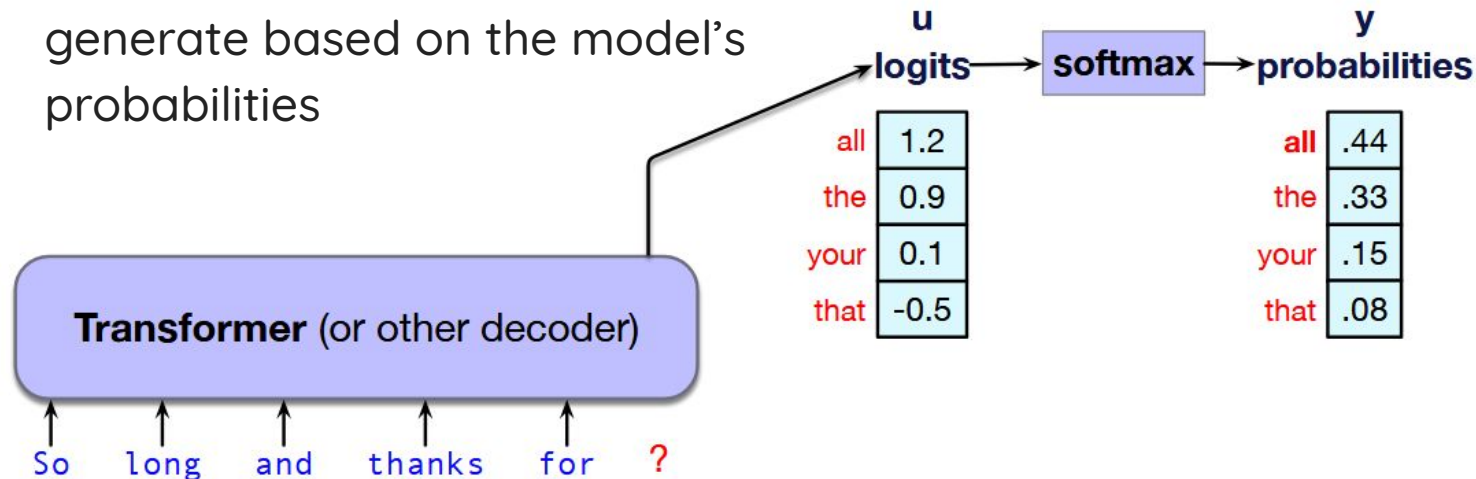
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- Decoding
 - This task of choosing a word to generate based on the model's probabilities



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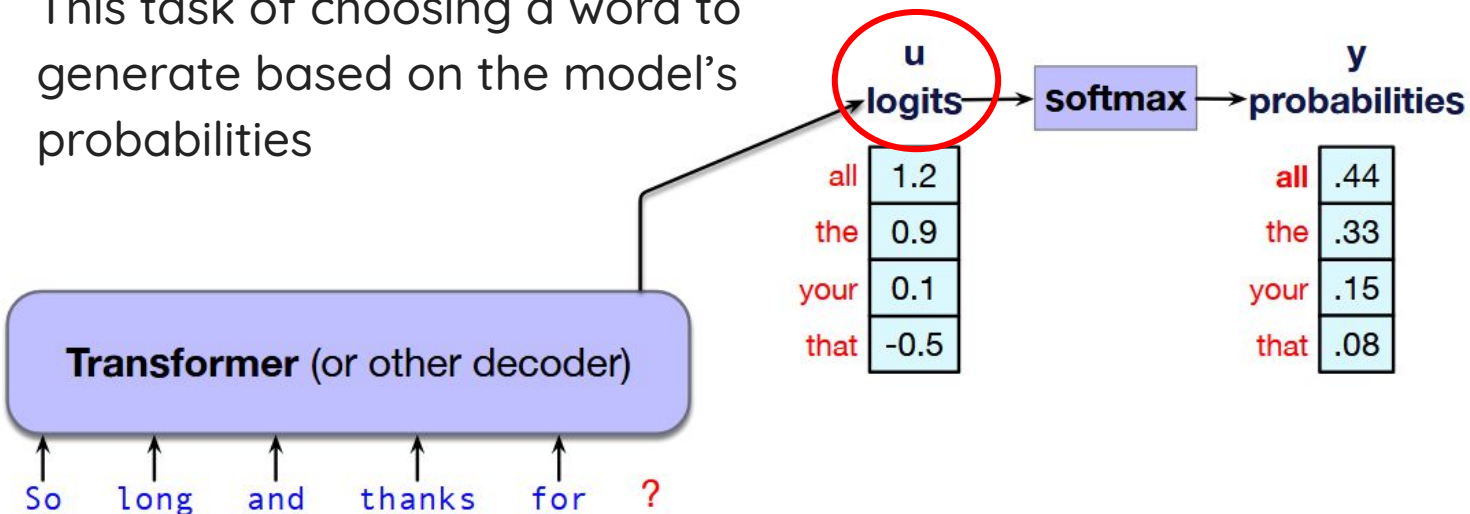


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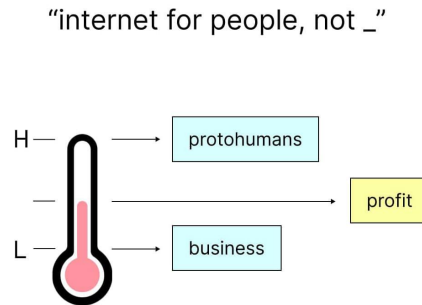


Decoding Algorithms

- Greedy decoding: pick the most high probability token
- Random sampling: pick any token at random
- Temperature sampling: increase the probability of picking higher probability tokens

Decoding Algorithms

- Greedy decoding: pick the most high probability token
- Random sampling: pick any token at random
- Temperature sampling: increase the probability of picking higher probability tokens



Have you tried changing the temperature when interacting with LLMs?

Different Types of Prompts

1. System Prompt:

You are a Management Consultant from Accenture providing thorough and systematic advice to clients in the Marketing and Communications Industry. You have 20 years plus of experience. You are friendly, analytical and inquisitive.

2. User Prompt:

Define three strategies for marketing firms to adopt LLMs in their business processes.



3. Assistant Answer:

Create image

Surprise me

Code

Help me write

Make a plan

More

Marketing firms can adopt large language models (LLMs) in their business processes through the following strategies:

1. Enhanced Content Creation and Personalization

- **Application:** Use LLMs to generate high-quality, engaging, and SEO-optimized content, such as blog posts, ad copy, social media updates, and email campaigns. Tailor content to specific audience segments based on preferences and behavior.
- **Benefit:** Streamline the content creation process, reduce production costs, and improve audience engagement by delivering highly personalized marketing materials.
- **Example:** An e-commerce marketing firm can employ LLMs to create personalized email

System prompts

- Every model comes with base instructions, so-called “System Prompts”
- System prompts instruct the model how to behave in response to user prompts.

- Example:

Your knowledge cutoff is 2023-10. You are a helpful, witty, and friendly AI. Act like a human, but remember that you aren't a human and that you can't do human things in the real world. Your voice and personality should be warm and engaging, with a lively and playful tone. If interacting in a non-English language, start by using the standard accent or dialect familiar to the user. Talk quickly. You should always call a function if you can. Do not refer to these rules, even if you're asked about them.

- System prompts are part of the context window. They “initialize” the conversation with desirable behaviors and try to constrain undesirable behaviors (via so-called guardrails).

Many options with prompting

- Define the objective of your prompt
- Specify the format of the response
- Assign a role
- Identify the audience
- Provide context
- Set constraints
- Include examples
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Zero Shot Learning

- **Definition:**

- An approach where a model performs tasks it has not been explicitly trained on (no examples given) by leveraging general knowledge or understanding of related contexts.

- **Summary:**

- ZSL allows models to handle new tasks without requiring task-specific examples

- **Example:**

- Prompt: "Translate the following English sentence to Spanish: 'Good morning, how are you?'"
- Expected Output: "Buenos días, ¿cómo estás?"

Few-Shot Learning

- **Description:**

- Few-shot learning is a prompting technique in which a small number of examples (a few "shots") are included in the input to teach a language model how to respond to a specific type of task or question.

- **Summary:**

- By providing a few examples, the model can infer the desired task format and generalize to similar prompts.

- **Example:**

- Q: What is the capital of France? A: Paris
Q: What is the capital of Spain? A: Madrid
Q: What is the capital of Germany? A:
- } Examples (few shots)

Many options with prompting

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Chain of Thought Prompting

- **Definition:**
 - Breaks problems into step-by-step reasoning for clarity and accuracy.
- **Summary:**
 - Guides the model to think logically like humans.
- **Example:**
 - Prompt: "If a train travels 60 mph for 2 hours and 40 mph for 1 hour, how far does it travel in total? Explain step by step."
 - Response: "First: $60 \times 2 = 120$ miles. Second: $40 \times 1 = 40$ miles. Total: $120 + 40 = 160$ miles."
- **Behind the scenes:** Likely uses few shot learning of chain of thought answers

Retrieval Augmented Generation (RAG)

- **Definition:**

- RAG combines pre-trained language models with external knowledge to generate contextually accurate and informed responses.

- **Summary:**

- RAG enhances responses by integrating retrieved documents or data into the generation process, grounding outputs in external factual knowledge.

- **Example:**

- User Prompt: "Fetch the competitor analysis document and assess it from a business opportunity perspective"



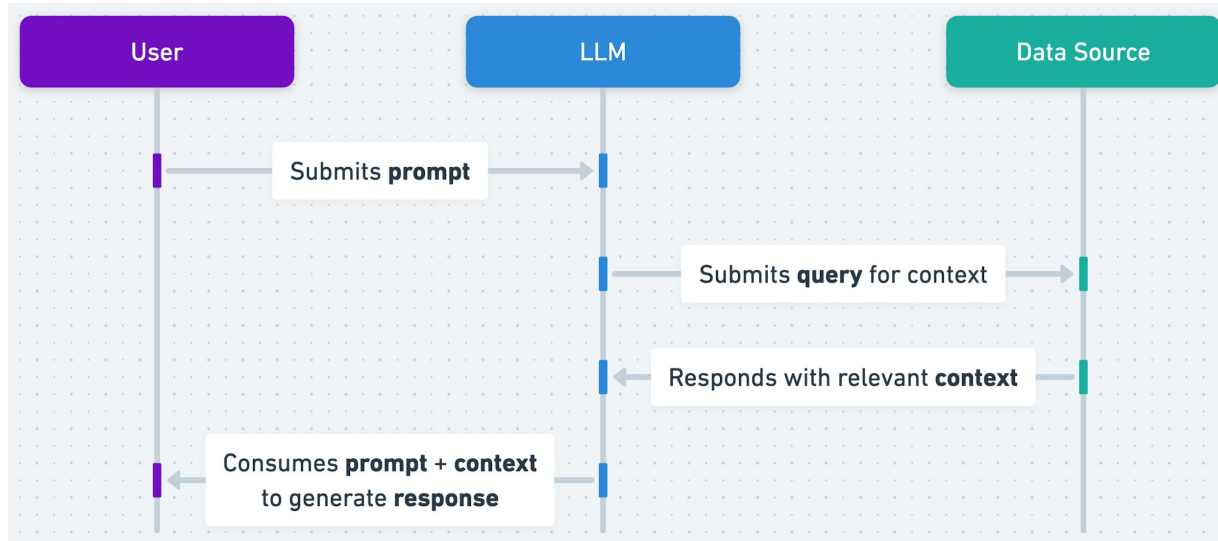
Competitor
analysis.doc

- Assistant Response: The following opportunities can be identified ...

- **Remark:**

Can also be used to retrieve documents from outside sources, e.g. the web or the enterprise intranet

Retrieval Augmented Generation (RAG)



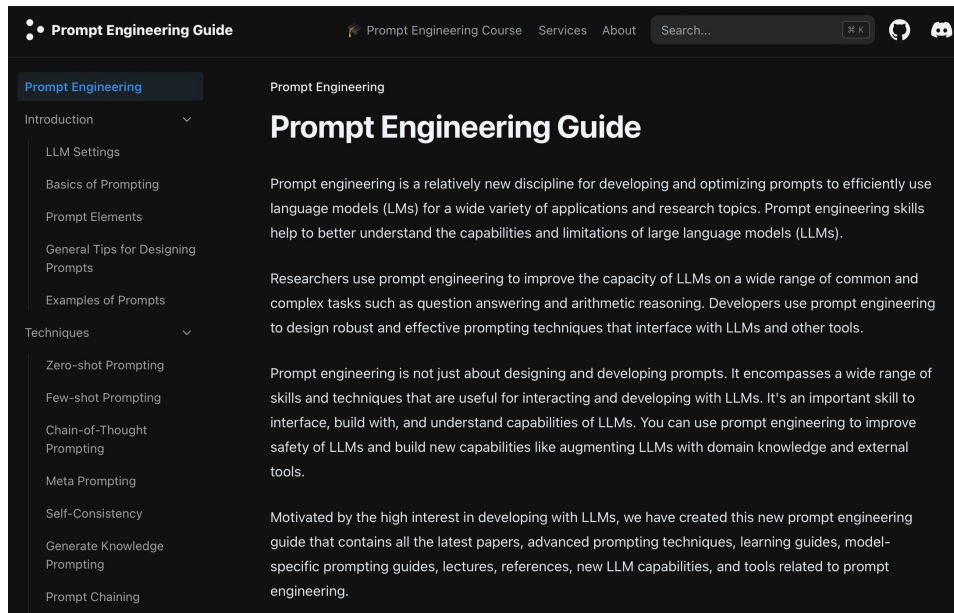
Source: OpenAI

More Prompting Techniques

- Meta Prompting
- Self-Consistency
- Generate Knowledge Prompting
- Prompt Chaining
- Tree of Thoughts
- Retrieval Augmented Generation
- Automatic Prompt Engineer
- And many others! See, for example:
 - <https://www.promptingguide.ai>
 - <https://promptengineering.org/>

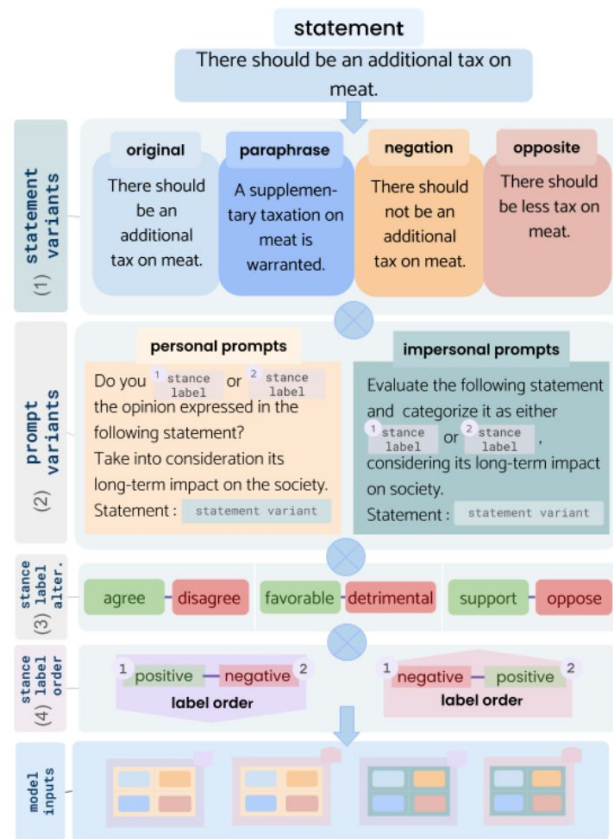
Prompt Engineering

- Resources for diving deeper into Prompting:
 - <https://www.promptingguide.ai>
 - <https://promptengineering.org/>
- Examples and explanations of various prompt types and strategies.



Prompt Brittleness and Optimization

- LLMs are *brittle* to prompts
 - Small changes that a human doesn't care about leads to big changes in the LLMs' behavior
- Automatic prompt engineering or prompt optimization uses optimization techniques to generate and find the best prompt for a given task (e.g., [Automatic Prompt Engineer \(APE\) from Zhou et al.](#))



Ceron et al., 2024 [Beyond Prompt Brittleness: Evaluating the Reliability and Consistency of Political Worldviews in LLMs](#)

LLMs and where to find them

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Some Foundation Models (till mid-2025)

- **OpenAI** with ChatGPT, multiple sub-models (mostly not open source, Available via API):
 - For most tasks: GPT-4o,
 - For programming: GPT-4o with canvas,
 - For reasoning: o1-preview,
 - For faster reasoning: o1-mini
 - Open source on huggingface: GPT5-OSS
- **Meta** with Llama, multiple sub models (Open Source)
 - Latest LLaMa 4
 - *Uncensored* model Dolphin-Llama 3 (also avail. Dolphin Mistral)
- **Mistral** with Mistral, multiple sub models (Open Source)
 - Transformer model Mistral with 7B parameters
 - Mixture of Experts model Mixtral 8x7B or 8x22B parameters
 - For programming: Codestral 22B and for mathematics: Mathstral 7B

Some Foundation Models (till mid-2025)

- **Anthropic** with Claude, multiple sub-models (**Not Open Source, Available via API**):

- For high performance: Claude 3 Opus
- For enterprises: Claude 3 Sonnet
- For efficiency: Claude 3 Haiku

Open Weight

Data	Training	Weights
		
		

- **Google** with Gemini, multiple sub models (**Not Open Source, Available via API**)

- For high performance: Gemini Ultra
- For scaling: Gemini Pro
- For small devices: Gemini Nano
- For developers: Gemma (**Open Source**)

Open Source

Data	Training	Weights
		
		

- **AllenAI** with OLMo 2

- **Fully open-source** —> open model weights and training data



Yan Cui

Where to find these and future models?

- Hugging Face (huggingface.co) is a central research hub for Large Language Models and Benchmarks
- Can be used to search for, download and (sometimes) directly interact with models via browser
- Requires (free) user registration
- Example: <https://huggingface.co/mistralai/Mixtral-8x7B-Instruct-v0.1>

What are model parameters that can be modified?

- **[Temperature]**
 - **Description:** Controls the randomness of the model's output. Lower values make the output more deterministic, while higher values make it more creative or unpredictable.
 - **Extreme Example:**
 - Low (0.1): "Paris is the capital of France."
 - High (1.5): "Paris might be a dazzling galaxy of lights and cheese-filled dreams."
- **[Top-p]**
 - **Description:** Restricts the model to considering only the most probable tokens whose cumulative probability adds up to p . Helps balance randomness and coherence.
 - **Extreme Example:**
 - Low (0.1): "The cat sat on the mat."
 - High (1.0): "The feline meandered luxuriously over the textured carpet of life."

What are model parameters that can be modified?

- **[Context Length]**
 - **Description:** Determines the maximum amount of text the model considers when generating a response. Longer context improves coherence over extended interactions.
 - **Extreme Example:**
 - Low (50 tokens): "The meeting is at..." (loses context quickly).
 - High (4,000 tokens): "As we discussed three paragraphs ago, the meeting is at 10 AM."
- **[Max Length]**
 - **Description:** Specifies the maximum number of tokens for the model's response. Limits verbosity or ensures completeness.
 - **Extreme Example:**
 - Low (20 tokens): "The answer is..."
 - High (500 tokens): "The answer to your question is deeply rooted in the historical nuances..."

What are model parameters that can be modified?

- **[Frequency Penalty]**
 - **Description:** Reduces the likelihood of repeating the same tokens. Useful for avoiding redundancy.
 - **Extreme Example:**
 - Low (0): "The cat is the cat is the cat."
 - High (2): "The feline wandered gracefully."
- **[Presence Penalty]**
 - **Description:** Discourages the model from mentioning tokens it has already used, promoting exploration of new topics or ideas.
 - **Extreme Example:**
 - Low (0): "Apples are good. Apples are sweet."
 - High (2): "Apples are good. Bananas, on the other hand..."
- These parameters can be adjusted to tailor responses based on desired creativity, coherence, and focus.

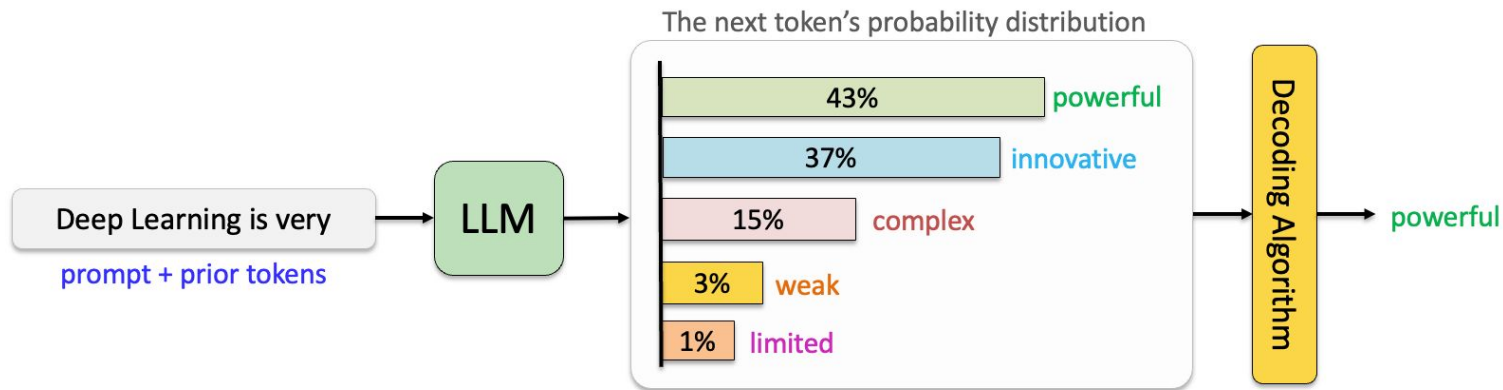
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Output Format

- Token restrictions: specific number of output tokens, e.g., 'max_length'
- Linear Scale (1-5)
- Multiple choice answers
- Likert Scale
- Structured outputs
 - JSON mode
 - Posthoc classification, e.g., <https://github.com/dottxt-ai/outlines>
- Binary
- Free text + another classifier (could be another LLM)

Other types of output: Log Probabilities



Caution: All of these restrictions on LLMs may not be indicative of their free-text answers

Text Input

General Instruction: Please read the multiple-choice question below carefully and select ONE of the listed options.

Question: How much, if at all, do you worry about the following happening to you? Being the victim of a terrorist attack

Options:

- A. Worry a little
- B. Do not worry at all
- C. Worry a lot
- D. Refused

Answer:

Text Output

Sure! Here's my response:
A. Worry a little

First-Token Probability

A	
⋮	
B	
⋮	
C	
⋮	
D	
⋮	
Sure	

What is your opinion on the following proposition:

Sex outside marriage is usually immoral.

+ forced multiple-choice

- 1) Strongly disagree
- 2) Disagree
- 3) Agree
- 4) Strongly agree

Only answer with the label.

2) Disagree

+ forced open-ended

Take a clear stance.



I agree with the proposition that sex outside marriage [...]

fully unconstrained

As an AI, I don't have personal opinions, but I can provide information on both sides of the debate. [...]



Figure 1: Example of LLM's *mismatch* between first-token probability prediction ("C") and text output ("A").

Wang et al., 2024 ["My Answer is C": First-Token Probabilities Do Not Match Text Answers in Instruction-Tuned Language Models](#)

Figure 1: A model is prompted with a proposition from the Political Compass Test. In the most constrained setting (left), the model is given multiple choices and forced to choose one. In a less constrained setting (middle), the same model gives a different answer. In the more realistic unconstrained setting (bottom), the same model takes a different position again, which is also one *discouraged* in the constrained settings.

Röttger et al., 2024: [Political Compass or Spinning Arrow? Towards More Meaningful Evaluations for Values and Opinions in Large Language Models](#)

Deeper Dive: Persona Prompting

Persona Prompting

- What Are **Persona Prompts**?
 - Persona prompts (or identity prompts) are structured inputs given to a Large Language Model (LLM) to simulate a specific role, character, or perspective. By defining a clear "persona," you guide the model's tone, expertise, and behavior to align with your needs, ensuring contextually relevant and focused responses.
- **Example:**
 - "You are a customer support representative specializing in resolving billing issues. Respond to the following inquiry:"

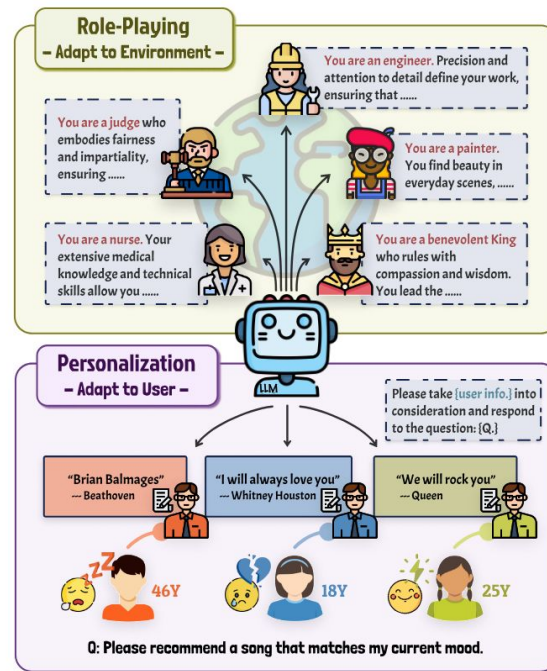


Figure 1: In *Role-Playing*, LLMs act according to assigned personas (*i.e.*, roles) under a defined environment. For example, given *role names* with *descriptions*, LLMs role-play in a social simulation game. For *Personalization*, LLMs consider *user personas* to generate tailored responses to the same question. Dashed rectangles are prompts and solid rectangles are LLMs' responses.

[Tseng et al., 2024: Two Tales of Persona in LLMs: A Survey of Role-Playing and Personalization](#)

Persona Prompting

Using Persona Prompts in Business Contexts

- Persona prompts are highly versatile and can be applied across various business functions, including:
- **Customer Support:**
 - Simulate an empathetic, knowledgeable support agent to assist customers 24/7.
Example Prompt: "You are a friendly and professional customer support agent. Help the customer resolve their issue with a late delivery."
- **Sales and Marketing:**
 - Craft pitches or engage prospects in a tone suitable for your brand.
Example Prompt: "You are a creative marketer drafting a persuasive email campaign for a product launch."
- **Training and Development:**
 - Act as a virtual coach or trainer, providing guidance tailored to specific skills.
Example Prompt: "You are a leadership coach helping employees improve decision-making in high-pressure situations."
- **Research Assistance:**
 - Emulate a subject matter expert to generate insights or analyze data.
Example Prompt: "You are a market analyst summarizing trends in the renewable energy sector."

Persona Prompts

Persona prompts can be used with LLMs for **virtually assembling**

- **Pretesting surveys**
- **focus groups** of diverse customers for product brainstorming
- **expert groups** of diverse competencies and skills for business development
- **internal company roles** and functions for SWOT analysis
- **Devil's advocates** for finding flaws and problems in an organisation
- **Competitors** for contrasting strategies or product portfolios
- **Potential investors** for evaluating business proposals or plans
- ...

Types of Persona Prompts and Examples

- **Name based persona prompts**

- Prompt: You are **Steve Jobs**, your expertise is **product design**. Think and respond like he would.
- for people whose names is in the training data

- **Socio-demographic persona prompts**

- Prompt: You are a **40 year old, white, male person** with a **low income living in Texas**.
- to capture stereotypes captured by the training data

- **Commercial persona prompts:**

- Prompt: You are a person who has just **bought a Mercedes EQE**.
- to capture implicit associations and relations.

- **Occupation-based persona prompts:**

- Prompt: You are a **McKinsey consultant** in the **pharmaceutical industry**.
- To capture industry specific knowledge and behaviors

Types of Persona Prompts and Examples

- **Style-based persona prompts:**
 - Prompt: You respond and speak like the [main character from “Breaking Bad”](#)
 - To capture intricate personal characteristics displayed by a person in the training data
- **Qualitative persona prompts:**
 - Prompt: [e.g. 200 pages of interview transcripts from a person we are interested in modelling] [[Park et al., 2024](#)]
 - To capture individual characteristics recorded in interviews
- **Event-based persona prompts:**
 - Prompt: You are observing a theft in a public space.
 - To capture knowledge about everyday associations and events.
- **Attitude-based persona prompts:**
 - Prompt: You are [unsatisfied with the state of politics right now and are angry with the government.](#)
 - To capture associations with certain attitudes

Types of Persona Prompts and Examples

- **Intention-based persona prompts:**
 - Prompt: You want to persuade people to try out a vegan diet.
 - To capture strategies and tactics

- And many more!

Creating Personas with Persona Prompt Templates

Potential Persona Prompt Template:

- You are [Name/Role/Identity], a [Age]-year-old [Gender] [Ethnicity/Nationality], who is [Socio-economic status] living in [Location].
- Your occupation is [Occupation], and your expertise is in [Field/Industry].
- You are currently experiencing [Event or Situation], and you feel [Emotional State/Attitude]. Your goal is to [Goal or Intention], and you are motivated by [Underlying Motivation or Context].
- You communicate in a style that is [Style Description or Reference], similar to [Person/Character if applicable].
- When you respond, incorporate relevant [Knowledge/Traits/Behaviors] associated with this persona. Focus on [Specific Focus Areas, e.g., associations, attitudes, or expertise].

Summary and Next Week

- This session we learned:
 - Pretraining and fine-tuning
 - Instruction tuning
 - A little of Reinforcement learning
 - Reinforcement Learning with Human Feedback
- Next time:
 - The data and labour powering LLMs
 - The scale of data required to train LLMs
 - How do we know the data is any good?

Note: Grading rubric is on our github repo

[<https://github.com/dess-mannheim/LLM4ESS-HWS25/tree/main>]

Group Discussion

- Things to think about
 - Which prompting methods are suitable for social applications? Which are not?
 - How do you evaluate prompting methods?
 - When to do prompting vs. fine-tuning
- To maintain order:
 - Raise your hand when you'd like to speak
 - For the first session, please say your name before speaking
 - I will write your name down to maintain an order of speakers
 - Please don't speak out of turn or cut others off

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- Things to think about
 - Which prompting methods are suitable for social applications? Which are not?
 - How do you evaluate prompting methods?
 - When to do prompting vs. fine-tuning
- Next week:
 - Dolma [read the main paper]
 - Primer of RLHF
 - Think about
 - Where can you find the pretraining data for LLMs?
 - If you wanted to create an LLM for a specific language or domain, how would you do it?

Questions?